

Microsoft
AI Agents Hackathon 2025

DashFlow

Streamlit Dashboard AI Agent

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Overview

Streamlit is an easy-to-use web app UI framework designed to help Data Analysts and Data Scientists present insights from their work seamlessly. It enables rapid prototyping and interactive dashboard creation with just a few lines of Python code. Analysts often spend hours transforming raw data into meaningful visuals, and Streamlit makes sharing those insights effortless.

Streamlit:

- Simple Python scripting for UI creation
- Built-in support for charts, maps, and media
- Fast deployment and sharing of dashboards
- Live data updates with caching and reruns
- Supports custom widgets and integrations

DashFlow, on the other hand, is an AI agentic tool built for deeper automation and intelligence. It allows users to integrate with any database and generate contextual insights based on natural language queries. DashFlow accelerates decision-making by eliminating the need for manual data querying or report generation.

DashFlow:

- Natural language query interface
 - Smart database connectors (SQL, NoSQL, cloud DBs)
 - AI-driven insight generation and summarization
 - Embeddable and customizable agent workflows
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DashFlow Features

- **Natural Language to Dashboard**
Converts user queries like *"Show sales trends for the last 6 months"* into fully functional Streamlit dashboards.
- **Smart Data Integration**
Connects seamlessly with various database stores such as PostgreSQL, MySQL and SQLite
- **Auto-generated Visuals**
Dynamically creates appropriate charts (bar, line, pie, scatter, etc.) based on the intent of the query.
- **AI-Powered Data Insights**
Not only visualizes data but also summarizes key insights, anomalies, or trends using language generation.
- **Streamlit Code Export**
Gives users the ability to download or edit the generated Python code for further customization



Architecture Workflow

